# VAND 4.0 Industrial Track

Technical Guidelines

> ⚠ **Important**
>
> This document is a high-level overview of the VAND 4.0 Industrial Track technical guidelines.
>
> For official dates, deadlines and precise rule and prize eligibility specifications, please refer to the official rules document available on the main challenge website, which takes precedence over this document.

Participants will develop anomaly detection models that are robust to external variations and adaptable to real-world conditions. While many existing models are trained on normal images and evaluated using both normal and abnormal samples, they often fail to generalize effectively in practical deployments. In real-world environments, **data drift** frequently occurs due to changes in camera angles, **lighting conditions**, background noise, and other external factors. These variations can significantly degrade model performance. Moreover, **computational efficiency** is a crucial requirement for deploying anomaly detection systems in real-world settings. Modern production lines often demand visual inspection to operate under strict **real-time conditions** - even when running on **resource-constrained edge devices**. Both **robustness to changing lighting conditions** and **strong computational efficiency** form the foundation of this year's VAND 4.0 Industrial Track.

## Dataset

Participants will use the MVTec Anomaly Detection 2 (MVTec AD 2) dataset. MVTec AD 2 is a public anomaly detection benchmark dataset that follows the design of previous popular anomaly detection datasets like MVTec AD or VisA. In particular, it contains anomaly-free images for training and validation and both anomaly-free and anomalous images for testing.

However, MVTec AD 2 aims to bridge academic research with industrial requirements in two ways. First, it contains 8 new challenging real-world scenarios captured under varying lighting conditions to reflect real-world distribution shifts. Second, the ground truth of the official test set is non-public to emphasize the unsupervised nature of industrial anomaly detection, i.e., not knowing which defects to expect at inference time. For development purposes, a small set of normal and anomalous test images with public ground truth is included in the dataset download.

For more information on MVTec AD 2 please refer to the MVTec AD 2 dataset paper:

Lars Heckler-Kram, Jan-Hendrik Neudeck, Ulla Scheler, Rebecca König, Carsten Steger: The MVTec AD 2 Dataset: Advanced Scenarios for Unsupervised Anomaly Detection; in: *International Journal of Computer Vision* 134(4), 2026, DOI:10.1007/s11263-026-02743-0

# Evaluation

This year's VAND 4.0 Industrial track will evaluate both general model performance and computational efficiency. Especially in modern industrial manufacturing, time and resources are critical constraints. Deployment environments often provide limited compute power and memory, making efficiency a key requirement alongside accuracy. As a result, even high-performing models may not be suitable for real-world use if they fail to meet practical efficiency standards.

To foster innovation aligned with common industrial hardware requirements, **the Industrial Track will not only reward high detection performance but also computational efficiency**. Detailed information on how these criteria are measured can be found in the sections below.

Prizes will be awarded to the two best overall-performing methods based on general model performance, the best zero-shot method as well as to the most efficient model that still achieves an adequate level of detection performance (also see "Prize Categories").

## General Model Performance

Models are validated and tested on a mix of normal and abnormal images to assess their anomaly detection capabilities. The focus is on enabling these models to effectively identify deviations from normality, emphasizing the real-world applicability of the techniques developed. In general, the evaluation protocol of the MVTec AD 2 dataset is followed.

General performance is evaluated using the pixel-level F1 scores (SegF1). This approach ensures a balanced consideration of precision and recall in the models' anomaly segmentation performance. Since anomaly maps are typically continuous, this evaluation also requires selecting a single decision threshold – a challenge often not yet considered within the scientific community but indispensable for deployment in real-world applications.

The final metric to assess model performance in the Industrial Track of the VAND 2026 Challenge considers the overall detection performance as well as robustness to real-world distribution shifts. It is calculated as the average rank of a model on the *private* and *private_mixed* test set in terms of the mean SegF1 over all eight object categories of MVTec AD 2:

**model_rank = (rank(SegF1['private']_average) + rank(SegF1['private_mixed']_average)) / 2**

where

**SegF1['private']** is the model performance on the test set that contains normal and anomalous images captured under the same lighting conditions as the training images (private test set)

**SegF1['private_mixed']** is the model performance on the test set that contains normal and anomalous images captured under a variety of lighting conditions both seen and unseen in the training images (private_mixed test set)

In case of an equal average rank, the method with a lower absolute difference between performance on the private and private_mixed test set is favoured.

## Computational Efficiency

Computational efficiency will be measured **after the submission deadline** and evaluated as a composite score across four metrics: power consumption, memory footprint, inference throughput, and performance (performance is measured as SegF1_average = (SegF1['private']_average + SegF1['private_mixed']_average) / 2). As for the other prizes, it is of paramount importance that your submission includes a self-contained, reproducible codebase and any pre-trained weights needed to replicate results. Please also refer to the official rules for further information.

# Model Design and Data Usage

In general, models are required to be designed class-agnostic, i.e., the **hyperparameters and overall architecture must stay the same for all object categories or automatically adapted to the object to be inspected**. For example, it is admissible to derive a threshold based on the validation data through an automated process but is not allowed to select "PatchCore" for one object category and "EfficientAD" for another object category to reach high performance. When in doubt about what is admissible, participants are advised to reach out to the organizers for clarification before the challenge deadline.

Besides, model design and training protocol must follow the requirements of the Regular Setting or the Zero-Shot Setting outlined below, depending on the track scenario participants want to take part in.

## Regular Setting

Participants are required to develop models based on the one-class training paradigm, which is **training exclusively on normal images of the object category to be evaluated**.

Consequently, images of the train split of MVTec AD 2 can be used for training and validation images can be used for hyperparameter estimation. It is not allowed to use any images of the public test split of MVTec AD 2 for training. Pre-training a model on anomalous images from other objects than the one to be evaluated or data from other publicly available datasets is admissible.

Training might follow the unified approach, i.e., training on normal data from all object categories, but evaluation metrics will be computed for each object category individually in line with the standard MVTec AD 2 evaluation protocol.

## Zero-Shot Setting

Models designed for the Zero-Shot Sub-Track may use **any publicly available data during training but no data from the MVTec AD 2 dataset** (including data from other object categories than the one to be evaluated).

# Submission Platform: [MVTec Benchmark Server](https://benchmark.mvtec.com/)

The MVTec Benchmark Server ([https://benchmark.mvtec.com/](https://benchmark.mvtec.com/)) serves as the official submission platform for the Industrial Track of the VAND 2026 Challenge. Leaderboards for the regular setting, zero-shot setting and overall best performance are available under [https://benchmark.mvtec.com/vand2026-leaderboard](https://benchmark.mvtec.com/vand2026-leaderboard). The leaderboard for model efficiency will be generated after the submission deadline.

## Submission Process

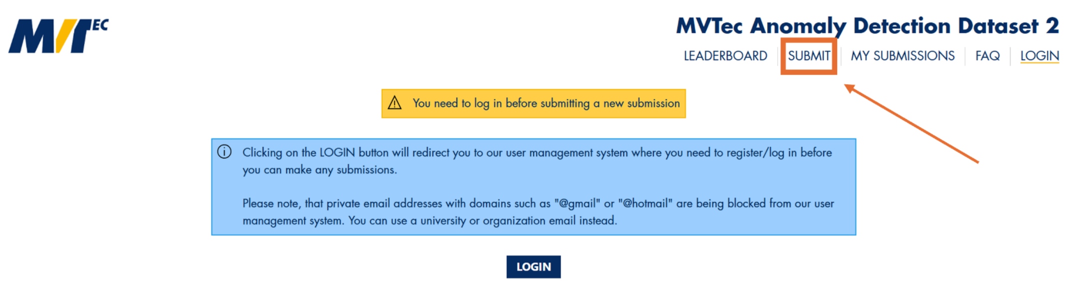
To submit to the Industrial Track of the VAND 2026 Challenge, you need to upload your model predictions (anomaly images + thresholded anomaly images) in the following way

1. Navigate to the 'SUBMIT' section of the MVTec Benchmark Server ([https://benchmark.mvtec.com/submit](https://benchmark.mvtec.com/submit)).
2. Click 'LOGIN' to login to the submission system/create an account.
3. Enter your university or organization email.
4. Login to an existing account /create a new account.
5. Follow the detailed submission instructions now visible to you.

6. Create a submission that meets all the requirements to participate in the Industrial Track of the VAND 2026 challenge (editing an evaluated submission is possible except for the uploaded data).

As soon as the evaluation has finished, your submission will be visible in the leaderboard (you can track the state of your submission under ‘my submissions’). More information can be found in the Frequently-Asked-Question (FAQ) section of the [MVTec Benchmark](MVTec Benchmark).

**Submitting to the _VAND 2026 Challenge Industrial Track_**
**on the MVTec Benchmark Server**




Navigate to the ‘SUBMIT’ section, login with your account or create an account, follow the detailed instructions now visible to you.
**IMPORTANT:** Consider the specific **SUBMISSION REQUIRMENTS** for the VAND 2026 Challenge Industrial Track.

## Submission Requirements

To participate in Industrial Track of the VAND 2026 challenge via the MVTec Benchmark Server:

1. **[R1]* Begin your method name by “VAND2026_{SETTING}”** followed by your personal method name, where SETTING is *either* “regular” or “zero_shot”, depending on the subcategory you want to enter:
    a. Example submission name for the regular setting: “VAND2026_regular MyMethodName”
    b. Example submission name for the zero-shot setting: “VAND2026_zero_shot MyMethodName”

2. **[R2] Include the continuous AND the thresholded anomaly images in your submission.** Otherwise segmentation quality can not be evaluated.

3. **[R3]* Open-source your code via specifying a 'Project link'.**
   You can find a template you can follow here: [https://github.com/cvpr-vand/vand-2026](https://github.com/cvpr-vand/vand-2026)
   Detailed requirements for the code submission:
   a. Share complete code to reproduce your results, including final model weights.
   b. Must be released under a non-commercial license (e.g., CC BY-NC 4.0)
   c. Hosting on established platforms (e.g., GitHub) is preferred.

4. **[R4]* Provide a link to a technical report** describing your method that is publicly accessible (e.g. arXiv) via 'Pub link'. You can also specify the title and the authors of your publication. Please follow the structure of the following template when creating your report: [VAND2026_challenge_report_template]()

**Please note that only submissions meeting all four criteria at the end of the challenge** (May 14th 29:59pm AOE) **will be considered as valid submissions** for the Industrial Track of the VAND 2026 Challenge.

* It is possible to edit the method name [R1] and to add/edit the link to the project [R3] and to the technical report [R4] after the successful evaluation of a submission. This also means you do not need to provide a link to the code and report until shortly before the submission deadline which allows you to keep your code and report confidential until then. After the submission deadline no modifications of existing entries will be possible and [benchmark.mvtec.com]() will undergo planned downtime for final evaluation of the challenge.

**Mandatory fields for the *VAND 2026 Challenge Industrial Track* when submitting to the MVTec Benchmark Server**

| | | |
|---|---|---|
| | Submission details | |
| **[R1]** | Method Name*: | **VAND2026_{SETTING} {MyMethodName}** |
| | Runtime in milliseconds: | *optional* |
| | Memory consumption in MB: | *optional* |
| **[R3]** | Project link (e.g. GitHub): | **https://github.com/{MyRepository}** |
| | Publication title: | *optional* |
| **[R4]** | Pub. link (e.g. Arxiv): | **Link to your PDF report** |
| | Authors: | *optional* |
| | File upload | |
| **[R2]** | Submission*: | **My_VAND2026_Challenge_Submission.tar.gz** |
| | **UPLOAD FILE** | |

Fields marked with * are required to create a submission (Method Name, Model Predictions).
Information highlighted in yellow can still be edited after evaluation (Method Name, Project Link, Publication Link).
The uploaded data cannot be modified any more. Please be aware of the evaluation budget and consider initial performance estimation on the public test set.

**EVALUATION BUDGET**

The evaluation budget per account is limited. This means that one participant is only allowed to make a certain number of submissions (= uploads) within a specific time. Currently, this limit is set to 3 submissions per week.

This setting avoids extensive hyperparameter tuning on the official test data of MVTec AD 2 (private and private_mixed test set). It highlights the concept of unsupervised anomaly detection, i.e., not knowing which defects and test data to expect. A small set of standard and anomalous test images with public ground truth is included in the dataset download (public test set) for development purposes.

We will freeze the VAND 2026 Industrial Track leaderboards at the end of the challenge (May 14th, 11:59 pm AOE) and disable new submissions and editing submissions until the evaluation of the Industrial track is completed. Submissions successfully uploaded by the end of the challenge will still be evaluated and considered for the final evaluation.

We will then filter for valid submissions according to the submission requirements, identify the best-performing methods and the most efficient model, and notify the winners via email.

# Prize Categories

Please also refer to the official rules for further information.

| *Category* | *Value* | *Description* |
|---|---|---|
| **Best performance** | $3000 USD | Submission with overall best **model_rank**. |
| **Best performance runner-up** | $750 USD | Submission with the overall second best **model_rank**. |
| **Best Efficiency** | ~$6000 USD | Submission with best efficiency/performance trade-off. |
| **Best performance Zero-Shot** | $3000 USD | Submission with overall best **model_rank** adhering to the zero-shot setting. |
| **Best paper** | ~$400 USD | Jury prize for an outstanding idea or methodological design. |